

Ali Burhan

Full Stack Developer

aliburhan.dev.ai@gmail.com | +92-300-1499488 | Lahore, Pakistan
linkedin.com/in/ali-burhan | github.com/Ali-Burhan | aliofficial.vercel.app

Professional Summary

Full Stack Developer with 3 years delivering production-grade cloud systems across enterprise safety, fintech, and AI verticals. Architected a microservices safety platform deployed across 1,000+ global sites. Independently built a \$10M+ business marketplace end-to-end in 4 months. Specializes in serverless AWS architectures, LLM integration (LangChain, RAG), and Infrastructure as Code with Terraform, with a track record of 99.9% uptime and 30% infrastructure cost reductions in production.

Technical Skills

Frontend: React, Next.js, TypeScript, Tailwind CSS, Zustand, React DnD

Backend: Python (FastAPI), Node.js (Express, NestJS), REST APIs, WebSockets, GraphQL

Databases: PostgreSQL, MongoDB, Redis, SQL Server

Cloud & Infrastructure: AWS (Lambda, API Gateway, S3, CloudFront, EC2, RDS, SQS, Kinesis, Lightsail), Terraform, Docker, Serverless Framework

AI / LLM: LangChain, OpenAI API, RAG, Prompt Engineering, Vector Embeddings, Agentic AI

DevOps & Observability: GitHub Actions, Jenkins, CI/CD, CloudWatch, Structured Logging, Jest, Pytest

Methodologies: Agile/Scrum, Jira, Sprint Planning, Cross-functional Collaboration

Professional Experience

HashLogics / Software Engineer

Lahore, Pakistan · Oct 2024 – Present

Forwood Safety — Client: Forwood Safety, Australia

Tech: Python, React, Terraform, AWS Lambda, API Gateway, SQS, Kinesis, Pytest

- Built and maintained a microservices safety platform across 1,000+ global sites using Python serverless architecture (AWS Lambda, API Gateway, SQS, Kinesis), reducing infrastructure costs by 30%.
- Automated infrastructure with Terraform, cutting deployment time from 2 hours to 15 minutes across dev, staging, and production environments.
- Implemented E2E testing with Pytest covering login flows, API validation, and regression checks; reduced post-release defects by 25%. Monitored production health via CloudWatch dashboards.
- Served as Release Manager coordinating quarterly releases and hotfixes via Jira and Jenkins, maintaining 99.9% uptime with zero-downtime deployments.

Lumaya — Client: Business Marketplace, Switzerland

Tech: Next.js, TypeScript, FastAPI, PostgreSQL, AWS Lightsail, WebSockets, Veriff KYC

- Sole developer, architected and delivered a full-stack B2B marketplace (lumaya.ch) in 4 months, handling 50+ listings totaling \$10M+ in value.
- Integrated Veriff KYC (100+ verifications, 95% success rate) and real-time WebSocket chat with file attachments and read receipts.
- Deployed on AWS Lightsail managing PostgreSQL optimization, SSL, and production environment at 99.8% uptime.

TechLing / Associate Software Engineer

Sheikhupura, Pakistan · May 2024 – Oct 2024

CazVid — Client: AI-Driven Job Platform, USA

Tech: Next.js, Zustand, NestJS, MongoDB, React DnD

- Built 25+ frontend components for an AI job board serving 150+ languages globally; integrated 20+ RESTful APIs for real-time AI resume screening and automated job matching.
- Developed a Kanban recruiter dashboard with React DnD managing 500+ postings; improved performance 40% via code splitting and lazy loading.

The Web Concept (TWC) / Junior Software Developer

Lahore, Pakistan · Aug 2023 – May 2024

Point of Sale & Web Applications — Internal Client Projects, Pakistan

Tech: PHP, JavaScript, React.js, Node.js, MongoDB, MySQL, REST APIs

- Built a full-featured POS system with inventory management, real-time transaction processing, receipt generation, and sales reporting, reducing manual billing errors by 40%.
- Transitioned legacy PHP UI components to React.js; integrated RESTful APIs with Node.js/MongoDB backend for dynamic product catalog and role-based access control.

Freelance Projects

Emotion-Aware Multilingual Voice Processing System (Jan – Apr 2024) — Client: Spain

Tech: FastAPI, Next.js, Deepgram API, OpenSmile, DeepL API, ElevenLabs API

- Built an end-to-end voice pipeline: audio capture, emotion detection via OpenSmile, STT with Deepgram, bidirectional EN↔ES translation via DeepL, and emotion-preserving TTS via ElevenLabs.

TeachTrack AI (Jun – Sep 2024) — Client: Turkey

Tech: Python, LangChain, OpenAI API, Next.js, RAG, Vector Embeddings

- AI-powered education platform automating test creation and grading for 1,200+ students and 45+ teachers across 3 schools; GPT-4 test generator reduced teacher prep time by 70%.
- Integrated LangChain RAG with vector embeddings for curriculum-based question generation from 1,000+ pages of course material.

Graduate Research

Papers written as part of MSc coursework at UET, not yet submitted for external publication.

Hallucination in Large Language Models: Causes, Detection, and Mitigation

Mar 2026

Systematic Literature Review · MSc CS, UET (Research Methodology Course)

Reviewed 7 papers (2024–2026) covering hallucination causes, detection signals (token-level entropy, self-consistency sampling, hidden-state probing), and mitigation strategies (Self-RAG, Chain-of-Verification, FT+RAG hybrid). Proposed a composite risk-scoring framework integrating three detection signals with a SHAP-based explainability layer absent from all reviewed works.

AI Self-Managing Operating Systems Architecture: A Comprehensive Survey

2025

Survey Paper · Co-authored with Ammar Shafiq · Supervised by Dr. Muhammad Aslam, UET

Surveyed AI/ML techniques, reinforcement learning, Deep Q-Networks, Proximal Policy Optimization (PPO), applied to OS resource management, scheduling, and fault tolerance. Compared architectures including Google Borg, DeepRM, and Kubernetes-based implementations across cloud, IoT, and mobile environments.

Agentic AI Systems (Ongoing)

2026

Graduate AI Coursework · MSc CS, UET

Exploring multi-agent frameworks, autonomous tool-use architectures, and agentic task-execution pipelines. Focus on LLM-based agents that can plan, act, and self-correct across complex workflows.

Certifications

- Python Essentials 1 & 2

Cisco

Education

University of Engineering and Technology (UET)

Lahore, Pakistan

Master of Science in Computer Science (In Progress) · Focus: ML, Distributed Systems, AI

2025 – 2027

NFC Institute of Engineering and Fertilizer Research

Faisalabad, Pakistan

Bachelor of Computer Science

Sep 2020 – Oct 2024